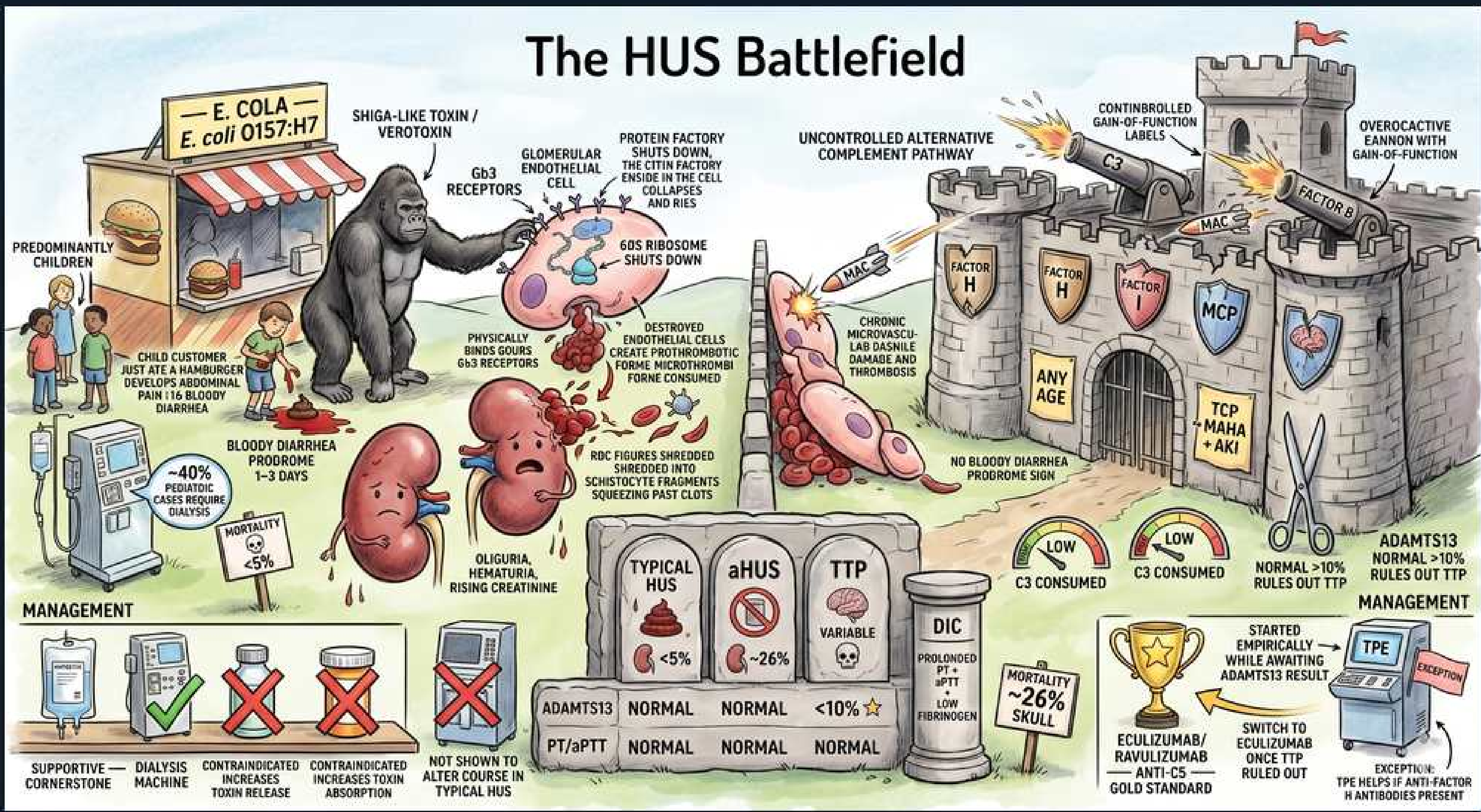
Platelet disorders — HUS (The HUS Battlefield)



1. E. coli O157:H7

WHAT YOU SEE

A burger stand sign reading 'E. COLA — E. coli O157:H7' — the undercooked-beef vendor pinning typical HUS on Shiga-toxin-producing E. coli O157:H7.

WHAT IT ENCODES

Typical (Shiga-toxin) HUS is most often caused by Shiga-toxin-producing E. coli (STEC), serotype O157:H7, acquired from undercooked ground beef, unpasteurized milk/juice, contaminated produce, or petting-zoo/farm exposure. Shigella dysenteriae can also produce Shiga toxin. STEC HUS is the most common cause of acute kidney injury in young children. The full syndrome is the triad: microangiopathic hemolytic anemia + thrombocytopenia + acute kidney injury.

BOARD TRAP

Don't anchor on a benign 'gastroenteritis' in a child with bloody diarrhea and order antibiotics — antibiotics in STEC colitis increase Shiga-toxin release and the risk of progressing to HUS. The right move is supportive care and monitoring renal function/CBC, not empiric antibiotics.

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2. Shiga toxin / verotoxin

WHAT YOU SEE

A hulking gorilla labeled SHIGA-LIKE TOXIN / VEROTOXIN — the brute-force toxin that shuts down the 60S ribosome and kills endothelium.

WHAT IT ENCODES

Shiga toxin (also called verotoxin / Shiga-like toxin) is the direct mediator of typical HUS. It is absorbed from the inflamed gut into the circulation and targets endothelium bearing the Gb3 (globotriaosylceramide) receptor — densely expressed in the renal glomerular microvasculature, hence the renal predominance. The toxin's A-subunit inactivates the 60S ribosomal subunit, halting protein synthesis and killing endothelial cells, which triggers microthrombi.

BOARD TRAP

Don't assume an antitoxin or antibiotic 'kills the toxin' — there is no effective antitoxin, and antibiotics that lyse bacteria can dump MORE toxin into the gut. Management of toxin-mediated HUS is supportive; you cannot neutralize the circulating Shiga toxin pharmacologically.

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3. Children

WHAT YOU SEE

A cluster of small child customers tagged 'PREDOMINANTLY CHILDREN' — typical STEC-HUS hits young kids (peak <5y), the leading cause of pediatric AKI.

WHAT IT ENCODES

Typical STEC-HUS predominantly affects young children (peak <5 years), usually 5–10 days after a diarrheal illness. It is the leading cause of acute kidney injury in this age group. Most children recover renal function with supportive care; a minority develop long-term CKD, hypertension, or proteinuria and need follow-up. Older age, adult onset, or absence of a diarrheal prodrome should raise suspicion for atypical (complement-mediated) HUS or TTP instead.

BOARD TRAP

Don't reflexively label every adult with MAHA + thrombocytopenia + renal failure as 'HUS' — in adults, TTP (severe ADAMTS13 deficiency) and atypical HUS are far more likely, and the management diverges sharply (plasma exchange for TTP, eculizumab for aHUS). Age and prodrome reshape the differential.

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4. Gb3 endothelial target

WHAT YOU SEE

The gorilla gripping Gb3 receptors on a glomerular endothelial cell whose ribosome 'shuts down' — Shiga toxin binds Gb3 (dense on renal endothelium) and halts protein synthesis.

WHAT IT ENCODES

Shiga toxin binds the Gb3 glycolipid receptor, which is highly concentrated on glomerular and other microvascular endothelial cells. Internalized toxin cleaves an adenine from 28S rRNA, inactivating the 60S ribosomal subunit and shutting down protein synthesis, causing endothelial cell death. The damaged, prothrombotic endothelium nucleates platelet-fibrin microthrombi in the renal microcirculation, producing the renal-dominant injury of typical HUS.

BOARD TRAP

Don't expect prominent neurologic involvement as in TTP — HUS targets Gb3-rich renal endothelium, so kidney injury dominates, whereas TTP (ADAMTS13 deficiency → ultralarge vWF multimers) more often causes fluctuating neurologic deficits. Using organ predominance to separate HUS from TTP is a classic discriminator.

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5. Bloody diarrhea prodrome

WHAT YOU SEE

A blood-streaked diarrhea prodrome panel marked '1-3 DAYS' before — the hemorrhagic-colitis warning that precedes typical HUS and steers you away from antibiotics.

WHAT IT ENCODES

Typical STEC-HUS is preceded by a prodrome of abdominal cramps and bloody diarrhea (hemorrhagic colitis), with HUS developing about 5–10 days after diarrhea onset (the sketch's 1–3 day window underscores the prodrome precedes the syndrome). The diarrheal history in a young child is the key clue that points to Shiga-toxin HUS rather than atypical/complement HUS or TTP, and it directs you AWAY from antibiotics.

BOARD TRAP

aHUS (complement) has NO bloody-diarrhea prodrome — its absence, plus any age and relapsing course or family history, should shift the diagnosis to atypical HUS (complement dysregulation) and prompt eculizumab rather than purely supportive care.

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6. AKI / renal dominant

WHAT YOU SEE

Sad, swollen kidney characters labeled 'OLIGURIA, HEMATURIA, RISING CREATININE' — kidney-dominant injury is the heart of the HUS triad.

WHAT IT ENCODES

Renal injury dominates the HUS picture: oliguria/anuria, hematuria and proteinuria, hypertension, and a rapidly rising creatinine (acute kidney injury). Severity ranges from mild creatinine elevation to dialysis-requiring failure. Supportive renal management — careful fluid/electrolyte balance, blood-pressure control, and dialysis when indicated — is the cornerstone of typical HUS care while the microangiopathy resolves.

BOARD TRAP

Don't over-resuscitate or volume-overload an oliguric HUS patient assuming 'pre-renal' AKI — the injury is intrinsic microangiopathic, and aggressive fluids in an anuric child cause pulmonary edema/hyperkalemia. Manage electrolytes and start dialysis for the standard indications rather than chasing urine output with fluids.

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7. Schistocytes MAHA

WHAT YOU SEE

Red cells being shredded into fragments as they squeeze past clots — schistocytes sheared by microthrombi, the MAHA of HUS.

WHAT IT ENCODES

HUS produces microangiopathic hemolytic anemia (MAHA): red cells are mechanically sheared into schistocytes (helmet/fragment cells) as they squeeze past the platelet-fibrin microthrombi lining damaged microvessels. Lab hallmarks are schistocytes on smear, elevated LDH, elevated indirect bilirubin, low haptoglobin, reticulocytosis, and a NEGATIVE direct Coombs (non-immune, mechanical hemolysis). The platelet count falls from consumption into microthrombi.

BOARD TRAP

Don't mistake MAHA for autoimmune hemolytic anemia — the Coombs test is NEGATIVE in HUS (mechanical destruction), whereas warm/cold AIHA is Coombs-positive. Treating HUS hemolysis with steroids as if it were AIHA misses the thrombotic microangiopathy entirely.

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8. Alternative complement (aHUS)

WHAT YOU SEE

A fortified castle labeled 'UNCONTROLLED ALTERNATIVE COMPLEMENT' with cannons firing — runaway alternative-complement activation driving atypical HUS.

WHAT IT ENCODES

Atypical HUS (aHUS) results from uncontrolled activation of the ALTERNATIVE complement pathway due to loss-of-function mutations in regulators (Factor H, Factor I, membrane cofactor protein/MCP/CD46) or gain-of-function mutations (C3, Factor B), or anti-Factor H antibodies. Unchecked C5/C5b-9 (membrane attack complex) activity injures endothelium and drives a chronic, relapsing thrombotic microangiopathy. It is NOT toxin-mediated and lacks a diarrheal prodrome.

BOARD TRAP

Don't treat aHUS as self-limited supportive-care HUS — without complement blockade (eculizumab/ravulizumab) it relapses and progresses to ESRD. Conversely, don't give eculizumab to straightforward STEC-HUS, which usually resolves with supportive care; matching mechanism to therapy is the test point.

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9. aHUS any age

WHAT YOU SEE

Castle banners reading 'ANY AGE' and 'NO BLOODY DIARRHEA' — aHUS strikes any age with no diarrheal prodrome, often relapsing.

WHAT IT ENCODES

aHUS occurs at ANY age (children and adults), often without a diarrheal prodrome, and may be triggered by infection, pregnancy, drugs, or transplant in a genetically susceptible person. It frequently relapses and may have a family history. C3 may be low (consumed) with normal C4 (alternative-pathway pattern); ADAMTS13 is normal. Diagnosis is largely clinical after excluding STEC-HUS and TTP, with genetic/complement testing to confirm.

BOARD TRAP

Don't be reassured by 'no diarrhea, normal coags' into calling it benign — that profile in an adult with MAHA + thrombocytopenia + AKI suggests aHUS (or TTP) and warrants urgent ADAMTS13 testing and consideration of empiric plasma exchange and complement blockade, not watchful waiting.

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10. Labs normal coags

WHAT YOU SEE

A lab bench panel showing 'ADAMTS13 NORMAL' and 'PT/aPTT NORMAL' — the fingerprint separating HUS from TTP (ADAMTS13 <10%) and DIC (deranged coags).

WHAT IT ENCODES

HUS labs: ADAMTS13 is NORMAL (>10%) and the coagulation studies PT/aPTT and fibrinogen are NORMAL — distinguishing HUS from DIC (prolonged PT/aPTT, low fibrinogen, high D-dimer) and from TTP (ADAMTS13 <10%). Expect anemia with schistocytes, thrombocytopenia, high LDH, low haptoglobin, negative Coombs, and renal markers (high creatinine, hematuria). The normal coags + normal ADAMTS13 with renal-dominant TMA is the HUS fingerprint.

BOARD TRAP

ADAMTS13 <10% = TTP, not HUS. Severely deficient ADAMTS13 redirects you to plasma exchange + steroids ± caplacizumab/rituximab for TTP; normal ADAMTS13 with renal predominance and a diarrheal prodrome points to STEC-HUS (supportive), and without prodrome to aHUS (eculizumab).

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11. Supportive care cornerstone

WHAT YOU SEE

A dialysis machine flagged 'SUPPORTIVE CARE = CORNERSTONE' with a green check — fluids, electrolytes, and dialysis as needed; typical HUS is self-limited.

WHAT IT ENCODES

Treatment of typical STEC-HUS is SUPPORTIVE — careful fluid and electrolyte management, transfusion of packed red cells for symptomatic anemia, blood-pressure control, and dialysis for the usual AKI indications (refractory hyperkalemia, acidosis, volume overload, uremia). The disease is largely self-limited as the toxin clears; most children recover. There is no proven role for routine plasma exchange in childhood STEC-HUS.

BOARD TRAP

PLEX is NOT first-line for typical childhood Shiga-toxin HUS — that's a TTP reflex. Plasma exchange is the answer for TTP (and is used for some aHUS scenarios pending eculizumab), but typical STEC-HUS is managed supportively; choosing PLEX here is a classic distractor.

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12. No antibiotics

WHAT YOU SEE

Antibiotics slashed with a red X captioned 'INCREASES TOXIN RELEASE' — antibiotics lyse STEC and dump more Shiga toxin, worsening HUS risk.

WHAT IT ENCODES

AVOID antibiotics in suspected STEC (O157:H7) infection — antibiotic-induced bacterial lysis increases Shiga-toxin release and may upregulate toxin production (SOS response), raising the risk of progression to HUS. Similarly avoid antimotility/antidiarrheal agents, which prolong toxin contact with the gut. Care for the colitis is supportive; reserve antibiotics for other documented infections, not STEC.

BOARD TRAP

Giving antibiotics for O157:H7 colitis is a classic wrong answer — a child with bloody diarrhea and suspected STEC should NOT get empiric antibiotics or antimotility drugs, as both increase HUS risk. The safe answer is supportive care with close monitoring of renal function and platelet count.

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13. Eculizumab for aHUS

WHAT YOU SEE

A gold trophy reading 'ECULIZUMAB / RAVULIZUMAB — ANTI-C5 — GOLD STANDARD' — terminal complement blockade, the game-changing therapy for aHUS.

WHAT IT ENCODES

Atypical (complement-mediated) HUS is treated with terminal complement blockade — eculizumab or longer-acting ravulizumab, monoclonal antibodies against C5 that prevent C5b-9/MAC formation — now the gold-standard therapy that has transformed outcomes. Because these agents block terminal complement, patients require meningococcal vaccination (and often prophylactic antibiotics) before/around initiation. Plasma exchange may be used as a temporizing measure while arranging eculizumab.

BOARD TRAP

Don't start a C5 inhibitor without meningococcal vaccination/prophylaxis — terminal complement blockade markedly raises the risk of Neisseria meningitidis infection. And don't withhold eculizumab in confirmed/strongly suspected aHUS waiting for genetics, since delay risks irreversible renal loss.

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14. Mortality / prognosis

WHAT YOU SEE

A mortality gauge swinging to the LOW end (<10%, '<5%') beside the DIC column — the favorable prognosis of typical childhood STEC-HUS.

WHAT IT ENCODES

Prognosis of typical childhood STEC-HUS is generally good: acute mortality is low (roughly <5%), and most children recover renal function with supportive care, though a minority develop long-term chronic kidney disease, hypertension, or proteinuria requiring follow-up. Atypical HUS has a worse natural history (high rates of ESRD and relapse) before/without complement blockade, underscoring why distinguishing the two matters.

BOARD TRAP

Don't equate the favorable prognosis of typical STEC-HUS with aHUS or TTP — untreated aHUS frequently progresses to end-stage renal disease and untreated TTP carries ~90% mortality. Assuming a benign, self-limited course for the wrong subtype delays the specific therapy that changes outcomes.
